

M.S. candidate at the University of Science and Technology of China with hands-on experience in PCB design, FPGA validation, mixed-signal IC development, and system debugging. Advanced a SPAD readout system from a board-level prototype to a custom 1x16-channel IC, spanning detector interfaces, PCB, Verilog/FPGA, transistor-level design, physical verification, and post-layout simulation.

EDUCATION

University of Science and Technology of China - M.S. in Quantum Science and TechnologySep 2024 - Jun 2027 (Expected)
Research: Semiconductor Single-Photon Detectors and Readout Circuits

Wuhan University - B.S. in PhysicsSep 2020 - Jun 2024
GPA: 3.82 / 4.00 | Comprehensive Rank: 4th in Major

TECHNICAL SKILLS

PCB & Hardware	PCB schematic and layout design, FPGA test boards, board bring-up, hardware debugging, oscilloscope and multimeter testing, system integration
EDA & IC Flow	Cadence Virtuoso (Schematic/Layout), Spectre/ADE, Calibre DRC/LVS/PEX, post-layout simulation, floorplanning, power distribution
Digital & Validation	Verilog, FPGA control and validation logic, CMOS digital logic, interface timing, mixed-signal system verification
Programming & AI	Python, C, YOLO-based computer vision, ROS communication and task-control workflows
Simulation	Ansys HFSS, 3D electromagnetic modeling, parameter sweeps, field-distribution analysis

RESEARCH & ENGINEERING EXPERIENCE

1x16-Channel SPAD Readout IC and Hardware Validation PlatformSep 2024 - Present

Core Designer | SMIC 180 nm BCD | Post-Layout Simulation and Pre-Tapeout Validation
PCB | FPGA | Verilog | Mixed-Signal IC | High-Voltage Isolation | IO/ESD | DRC/LVS/PEX

- Led the transition from a discrete board-level prototype to a custom 1x16-channel readout IC, covering board electronics, FPGA validation, chip architecture, transistor-level design, layout, and system verification.
- Designed the analog front end, active quenching and reset circuits, digital control logic, 1.8 V / 5 V domains, external 60 V quenching interface, high-voltage isolation, and IO/ESD protection.
- Completed chip floorplanning, power/ground distribution, guard-ring and substrate strategies, sensitive-signal isolation, DRC/LVS, parasitic extraction, and post-layout simulation using Virtuoso, Spectre, and Calibre.
- Designed an FPGA test-board schematic and PCB and developed Verilog-based control and validation logic, establishing an end-to-end detector-board-FPGA-IC verification platform.
- Redesigned the board-level active-quenching circuit and reduced measured average power from approximately 1 W to 0.1 W per channel.

INTERNSHIP EXPERIENCE

Baidu - Quantum Circuit Design and Electromagnetic Simulation InternSep 2023 - Dec 2023

- Built 3D electromagnetic models for superconducting quantum circuits and microwave structures in Ansys HFSS, including materials, boundaries, excitation ports, and mesh configurations.
- Performed parameter sweeps and field-distribution analysis to evaluate geometry effects and support layout and structural optimization.

ADDITIONAL PROJECT

Autonomous Ground Vehicle and UAV Collaborative Control SystemUndergraduate Project

Second Prize, Hubei Provincial College Student Electronic Design Competition

- Developed a Python/YOLO control workflow and ROS communication across perception, task-decision, and MCU motor-control modules; integrated the system into a validated closed loop.